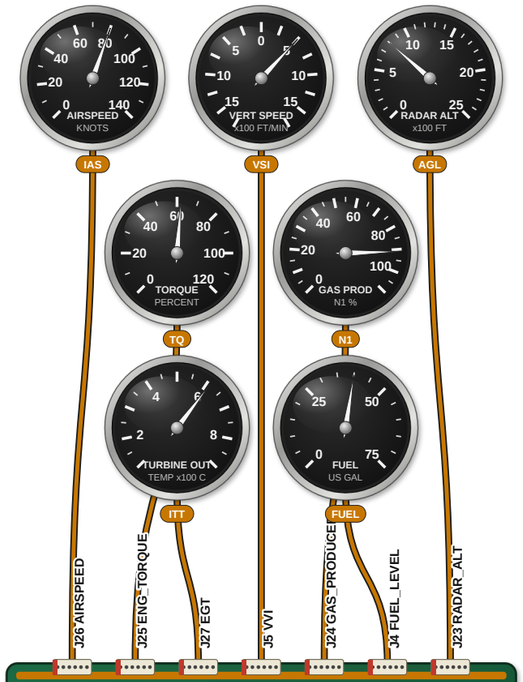
Jet Ranger cockpit: gauges, lamps and displays, and the controller board that drives each one

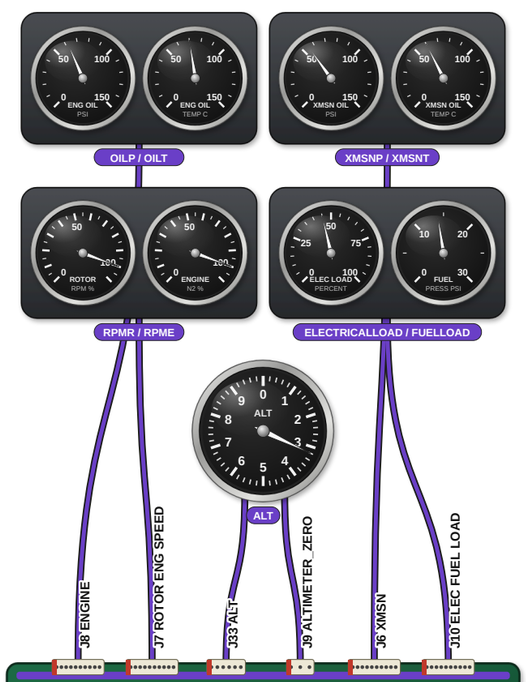
Instrument map, A3 landscape. Needles and lamps are drawn at illustrative readings. Cables end at the named connector on the board (references as printed on the PCBs).

STEPPER GAUGES (X27 steppers)



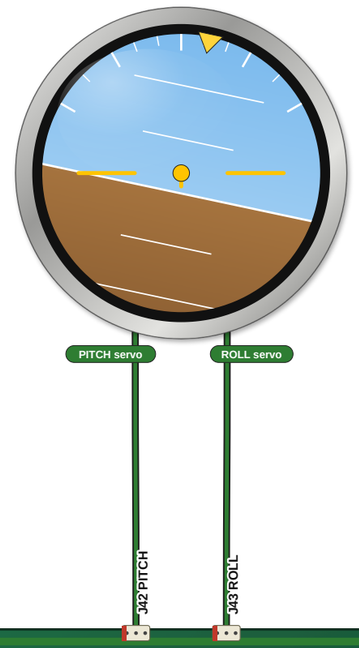
STEPPER CONTROLLER
172.16.1.105
JET_RANGER_STEPPER_CONTROLLER
Arduino Mega 2560 + W5500 | Jet Ranger 13 Stepper v1.6
ETHERNET 172.16.1.0/24

DUAL STEPPER GAUGES (dual instruments + altimeter)



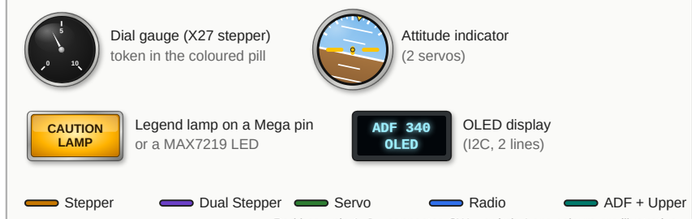
DUAL STEPPER CONTROLLER
172.16.1.106
JET_RANGER_OLED_DUAL_STEPPER_CONTROLLER
Arduino Mega 2560 + W5500 | OLED Dual Stepper v1.5
ETHERNET 172.16.1.0/24

ATTITUDE INDICATOR (pitch and bank servos)



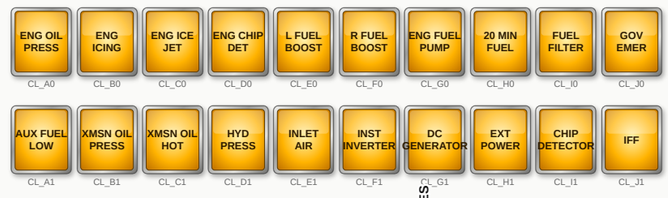
SERVO CONTROLLER
172.16.1.102
JET_RANGER_SERVO_CONTROLLER
Arduino Mega 2560 + W5500 | Front Panel Servo v1.4
ETHERNET 172.16.1.0/24

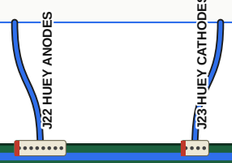
KEY



HUEY WARNING PANEL (20 caution lamps on MAX7219 digits 4 - 7)

connectors J20 - J23



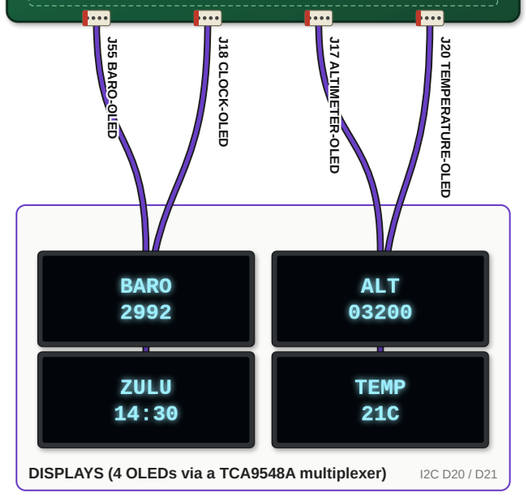


RADIO CONTROLLER
172.16.1.101
JET_RANGER_RADIO_CONTROLLER
Arduino Mega 2560 + W5500 | Radio Auto Pilot v1.4
ETHERNET 172.16.1.0/24

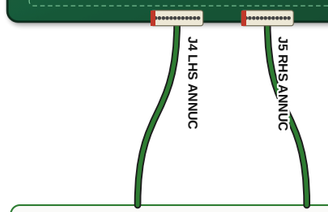
ADF PANEL

Switches, knobs, keypads and the USB HID board are inputs: not drawn.

ADF + UPPER
172.16.1.107
JET_RANGER_ADF_CONTROLLER
Arduino Mega 2560 + W5500 | Upper ADF v1.0
ETHERNET 172.16.1.0/24

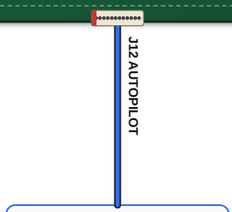


DISPLAYS (4 OLEDs via a TCA9548A multiplexer) I2C D20 / D21

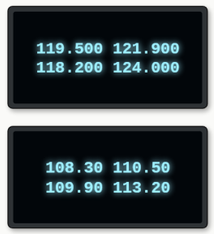


ANNUNCIATOR LAMPS (Mega pins A1 - A15)

ENGINE OUT (EOUT A1)	ROTOR LOW (RLOW A2)	XMSN OIL PRESS (TOPW A3)
XMSN OIL TEMP (TOTW A4)	BATTERY HOT (BHOT A5)	BATTERY TEMP (BTMP A6)
ENGINE CHIP (EC A7)	TAIL ROTOR CHIP (TRC A8)	XMSN CHIP (TC A9)
BAGGAGE DOOR (BD A10)	FUEL PUMP (FPMP A11)	AFT FUEL FILTER (FFLTR A12)
GEN FAIL (GENF A13)	LOW FUEL (LOWF A14)	SC FAIL (SCF A15)

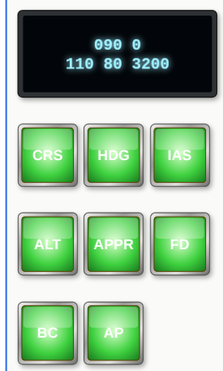


COM AND NAV PANEL



COM and NAV frequency OLEDs (8 encoders and 4 swap switches are inputs, not drawn)

AUTO PILOT PANEL 8 LEDs

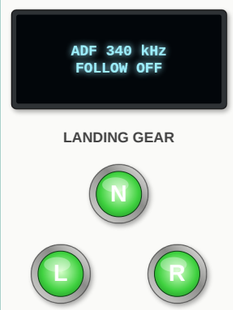


AUDIO PANEL 11 LEDs





ADF PANEL



HOW THE VALUES ARRIVE

SIMULATOR
MSFS 2020 / 2024, Prepar3D, DCS World

↓

BRIDGE PROGRAM (on the PC)
FSUIPC, MSFS or DCS-BIOS bridge

↓

ETHERNET 172.16.1.0/24
UDP 13136: D,IAS:93.4,PITCH:4.5,...

↓

CONTROLLER BOARD
Mega 2560 + W5500, calibration tables

↓

CABLE TO THE CONNECTOR
stepper coils, servos, lamps, OLEDs

Each board takes real units and does its own calibration.